

SOLIDWORKS API (.NET VBA C#) — Reference (50/337): IComponent2 (part 3 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IComponent2 (part 3 of 4) (Type)

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IListExternalFileReferences Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IListExternalFileReferences Method (IComponent2)

Obsolete. Superseded by [IComponent2::IListExternalFileReferences2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IListExternalFileReferences( _  
    ByVal NumRefs As System.Integer, _  
    ByRef ModelPathName As System.String, _  
    ByRef CompPathName As System.String, _  
    ByRef Feature As System.String, _  
    ByRef DataType As System.String, _  
    ByRef Status As System.Integer, _  
    ByRef RefEntity As System.String, _  
    ByRef FeatComp As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim NumRefs As System.Integer  
Dim ModelPathName As System.String  
Dim CompPathName As System.String  
Dim Feature As System.String  
Dim DataType As System.String  
Dim Status As System.Integer  
Dim RefEntity As System.String  
Dim FeatComp As System.String
```

```
instance.IListExternalFileReferences(NumRefs, ModelPathName, CompPathName, Feature, DataType,  
Status, RefEntity, FeatComp)
```

C#

```
void IListExternalFileReferences(  
    System.int NumRefs,  
    out System.string ModelPathName,  
    out System.string CompPathName,  
    out System.string Feature,  
    out System.string DataType,  
    out System.int Status,  
    out System.string RefEntity,  
    out System.string FeatComp  
)
```

C++/CLI

```
void IListExternalFileReferences(  
& System.int NumRefs,  
& [Out] System.String^ ModelPathName,  
& [Out] System.String^ CompPathName,  
& [Out] System.String^ Feature,  
& [Out] System.String^ DataType,  
& [Out] System.int Status,  
& [Out] System.String^ RefEntity,  
& [Out] System.String^ FeatComp  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumRefs
ModelPathName
CompPathName
Feature
DataType
Status
RefEntity
FeatComp

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

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IListExternalFileReferences2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IListExternalFileReferences2 Method (IComponent2)

Gets the names and statuses of the external references on the component.

.NET Syntax

Visual Basic (Declaration)

```
Sub IListExternalFileReferences2( _  
    ByVal NumRefs As System.Integer, _  
    ByRef ModelPathName As System.String, _  
    ByRef CompPathName As System.String, _  
    ByRef Feature As System.String, _  
    ByRef DataType As System.String, _  
    ByRef Status As System.Integer, _  
    ByRef RefEntity As System.String, _  
    ByRef FeatComp As System.String, _  
    ByRef ConfigOption As System.Integer, _  
    ByRef ConfigName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim NumRefs As System.Integer  
Dim ModelPathName As System.String  
Dim CompPathName As System.String  
Dim Feature As System.String  
Dim DataType As System.String  
Dim Status As System.Integer  
Dim RefEntity As System.String  
Dim FeatComp As System.String  
Dim ConfigOption As System.Integer  
Dim ConfigName As System.String
```

```
instance.IListExternalFileReferences2(NumRefs, ModelPathName, CompPathName, Feature, DataType,  
Status, RefEntity, FeatComp, ConfigOption, ConfigName)
```

C#

```
void IListExternalFileReferences2(  
    System.int NumRefs,  
    out System.string ModelPathName,  
    out System.string CompPathName,  
    out System.string Feature,  
    out System.string DataType,  
    out System.int Status,  
    out System.string RefEntity,
```

```

    out System.string FeatComp,
    out System.int ConfigOption,
    out System.string ConfigName
)

```

C++/CLI

```

void IListExternalFileReferences2(
& System.int NumRefs,
& [Out] System.String^ ModelPathName,
& [Out] System.String^ CompPathName,
& [Out] System.String^ Feature,
& [Out] System.String^ DataType,
& [Out] System.int Status,
& [Out] System.String^ RefEntity,
& [Out] System.String^ FeatComp,
& [Out] System.int ConfigOption,
& [Out] System.String^ ConfigName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumRefs

Number of external references

ModelPathName

- in-process, unmanaged C++: Pointer to an array of path names of documents of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

CompPathName

- in-process, unmanaged C++: Pointer to an array of path names of referenced components of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

Feature

- in-process, unmanaged C++: Pointer to an array of in-context items (sketches, features, and so on) of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

DataType

- in-process, unmanaged C++: Pointer to an array of data used to create the items (converted edge or face, converted or offset sketch entity, body, and so on) of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

Status

- in-process, unmanaged C++: Pointer to an array of the statuses of the external references as defined in [swExternalReferenceStatus_e](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

RefEntity

- in-process, unmanaged C++: Pointer to an array of actual items being used and the names of the documents that contain the items of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

FeatComp

- in-process, unmanaged C++: Pointer to an array of the names of the components in which the affected features exist of size NumRefs; this information is only displayed when one or more RefEntity is in a different component in an assembly and does not apply to derived parts
- VBA, VB.NET, C#, and C++/CLI: Not supported

ConfigOption

Configuration options as defined by [swExternalFileReferencesConfig_e](#)

ConfigName

Name of the configuration when configOption is swExternalFileReferencesNamedConfig

Remarks

Before calling this method, call [IComponent2::ListExternalFileReferencesCount](#) to specify NumRefs.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::ListExternalFileReferences2 Method](#)
[IComponent2::UpdateExternalFileReferences Method](#)
[IFeature::IListExternalFileReferences2 Method](#)
[IFeature::ListExternalFileReferences2 Method](#)
[IFeature::ListExternalFileReferencesCount Method](#)
[IFeature::UpdateExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IMaterialPropertyValues Property (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IMaterialPropertyValues Property (IComponent2)

Gets or sets the material properties for the selected component in the active configuration.

.NET Syntax

Visual Basic (Declaration)

```
Property IMaterialPropertyValues As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Double

instance.IMaterialPropertyValues = value

value = instance.IMaterialPropertyValues
```

C#

```
System.double IMaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.double IMaterialPropertyValues {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the material's values on this component (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::IMaterialPropertyValues](#).

Remarks

The material values include the component color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission. Valid values are from 0 to 1 for all values.

The format of the parameters or return values is an array of doubles as follows:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

To see a color change, you must:

1. Specify *R, G, and B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse, Specular, Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient, Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

This method returns an S_false HRESULT for COM implementations if the user has not explicitly modified this component from the material property values of the underlying part document. For example, the user can right-click a component in the FeatureManager design tree and select **Component Properties** to change the color. If the user has not done this, then this method returns NULL color information.

The default component color can be obtained from the component's [IModelDoc2](#) object ([IComponent2::IGetModelDoc](#)) using [IModelDoc2::MaterialPropertyValues](#).

You might also want to check for specific face colors using [IFace2::IMaterialPropertyValues](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::MaterialPropertyValues Property](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::IGetMaterialPropertyValues2 Method](#)
[IComponent2::IRemoveMaterialProperty2 Method](#)
[IComponent2::RemoveMaterialProperty2 Method](#)
[IComponent2::ISetMaterialPropertyValues2 Method](#)
[IComponent2::SetMaterialPropertyValues2 Method](#)

Availability

IRemoveMaterialProperty2 Method (IComponent2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IRemoveMaterialProperty2 Method (IComponent2)

Removes material property values from the component.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IRemoveMaterialProperty2( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim Config_opt As System.Integer
Dim Config_count As System.Integer
Dim Config_names As System.String
Dim value As System.Boolean

value = instance.IRemoveMaterialProperty2(Config_opt, Config_count, Config_names)
```

C#

```
System.bool IRemoveMaterialProperty2(
    System.int Config\_opt,
    System.int Config\_count,
    ref System.string Config\_names
)
```

C++/CLI

```
System.bool IRemoveMaterialProperty2(
    & System.int Config\_opt,
    & System.int Config\_count,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration options as defined by [swInConfigurationOpts_e](#)
Config_count
Number of configurations for this component
Config_names
Array of configuration names for this component

Return Value

True if the material property values are removed, false if not

Remarks

This method is intended to be used on components whose material property value has changed; for example, color.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::RemoveMaterialProperty2 Method](#)
[IComponent2::SetMaterialPropertyValues2 Method](#)
[IComponent2::SetMaterialPropertyValues2 Method](#)

[IComponent2::MaterialPropertyValues Property](#)
[IComponent2::IMaterialPropertyValues Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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IsDisplayDataOutOfDate Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsDisplayDataOutOfDate Method (IComponent2)

Gets the status of the display data for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IsDisplayDataOutOfDate() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer
```

```
value = instance.IsDisplayDataOutOfDate()
```

C#

```
System.int IsDisplayDataOutOfDate()
```

C++/CLI

```
System.int IsDisplayDataOutOfDate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Rebuild state of the components as defined in [swOutOfDateStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::IsDisplayDataOutOfDate](#).

Remarks

Tessellation information is available only when the component is loaded as lightweight.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IsEnvelope Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsEnvelope Method (IComponent2)

Gets whether this component is an envelope.

.NET Syntax

```
Function IsEnvelope() As System.Boolean
```



```
Dim instance As IComponent2
Dim value As System.Boolean

value = instance.IsEnvelope()
```

```
System.bool IsEnvelope()
```

```
System.bool IsEnvelope();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component is an envelope, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsEnvelope](#).

Example

[Change Component to Envelope \(C#\)](#)
[Change Component to Envelope \(VB.NET\)](#)
[Change Component to Envelope \(VBA\)](#)

Remarks

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CompConfigProperties5 Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ISetMaterialPropertyValues2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ISetMaterialPropertyValues2 Method (IComponent2)

Sets the material properties for this component.

.NET Syntax

```
Sub ISetMaterialPropertyValues2( _
    ByRef Material\_values As System.Double, _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
)
```



```
Dim instance As IComponent2
Dim Material_values As System.Double
Dim Config_opt As System.Integer
Dim Config_count As System.Integer
Dim Config_names As System.String
```

```
instance.ISetMaterialPropertyValues2(Material_values, Config_opt, Config_count, Config_names)
```

C#

```
void ISetMaterialPropertyValues2(
    ref System.double Material_values,
    System.int Config_opt,
    System.int Config_count,
    ref System.string Config_names
)
```

C++/CLI

```
void ISetMaterialPropertyValues2(
    & System.double% Material_values,
    & System.int Config_opt,
    & System.int Config_count,
    & System.String^% Config_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Material_values

Array of doubles of size 9 (see **Remarks**)

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_count

Number of configurations for this component

Config_names

- In-process, unmanaged C++: Pointer to an array of configuration names of size Config_count

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The material property values include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

The format of *Material_values* is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after calling this method.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::SetMaterialPropertyValues2 Method](#)

[IComponent2::GetMaterialPropertyValues2 Method](#)

[IComponent2::GetMaterialPropertyValues2 Method](#)

[IComponent2::RemoveMaterialProperty2 Method](#)

[IComponent2::RemoveMaterialProperty2 Method](#)

[IComponent2::IMaterialPropertyValues Property](#)

[IComponent2::MaterialPropertyValues Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

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ISetVisibility Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ISetVisibility Method (IComponent2)

Sets the visibility state for this component.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetVisibility( _  
    ByVal State As System.Integer, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim State As System.Integer  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
  
instance.ISetVisibility(State, Config_opt, Config_count, Config_names)
```

C#

```
void ISetVisibility(  
    System.int State,  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
void ISetVisibility(  
& System.int State,  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

State

Visibility state as defined in [swComponentVisibilityState_e](#)

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_count

Number of configurations for this component

Config_names

Array of configuration names of size Config_count

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::Visible Property](#)
[IComponent2::SetVisibility Method](#)
[IComponent2::GetVisibility Method](#)
[IComponent2::IsHidden Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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ISetXform Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ISetXform Method (IComponent2)

Obsolete. Superseded by [IComponent2::Transform2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISetXform(  
    ByRef XformIn As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim XformIn As System.Double  
Dim value As System.Boolean  
  
value = instance.ISetXform(XformIn)
```

C#

```
System.bool ISetXform(  
    ref System.double XformIn  
)
```

C++/CLI

```
System.bool ISetXform(  
& System.double% XformIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XformIn

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

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ISetXformAndSolve Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ISetXformAndSolve Method (IComponent2)

Obsolete. Superseded by [IComponent2::SetTransformAndSolve2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISetXformAndSolve(  
    ByRef XformIn As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim XformIn As System.Double  
Dim value As System.Boolean  
  
value = instance.ISetXformAndSolve(XformIn)
```

C#

```
System.bool ISetXformAndSolve(  
    ref System.double XformIn
```

```
)
```

C++/CLI

```
System.bool ISetXformAndSolve(  
& System.double% XformIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XformIn

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

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IsFixed Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsFixed Method (IComponent2)

Gets whether the component is fixed or floating.

.NET Syntax

Visual Basic (Declaration)

```
Function IsFixed() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.IsFixed()
```

C#

```
System.bool IsFixed()
```

C++/CLI

```
System.bool IsFixed();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component is fixed, false if it is floating

Visual Basic for Applications (VBA) Syntax

See [Component2::IsFixed](#).

Example

[Get Component by Name \(C#\)](#)
[Get Component by Name \(VB.NET\)](#)
[Get Component by Name \(VBA\)](#)
[Get Component State \(C#\)](#)
[Get Component State \(VB.NET\)](#)
[Get Component State \(VBA\)](#)

Remarks

This method only applies to the top level of components. To get the actual state of sub-assemblies, you must get the [IModelDoc2](#) object of the subassembly, show the desired configuration, and get the state (fixed or floating) of the lower components.

To determine if a component is fixed or floating, you must begin the traversal from the subassembly document in the appropriate configuration instead of from the root level. At the root level, all of the components in the subassembly are allowed to move.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::FixComponent Method \(\)](#)
[IAssemblyDoc::TemporaryFixGroup Method \(\)](#)
[IAssemblyDoc::TemporaryFixGroupExit Method \(\)](#)
[IAssemblyDoc::UnfixComponent Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IsGraphicsOnly Property (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsGraphicsOnly Property (IComponent2)

Gets whether this component is graphics only.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsGraphicsOnly As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.IsGraphicsOnly
```

C#

```
System.bool IsGraphicsOnly {get;}
```

C++/CLI

```
property System.bool IsGraphicsOnly {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if graphics only, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsGraphicsOnly](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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IsHidden Method (IComponent2)

[See Also](#) [Example](#)



Visual Basic Syntax

Visual Basic (Declaration)

```
Function IsHidden( _  
    ByVal ConsiderSuppressed As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IComponent2  
Dim ConsiderSuppressed As System.Boolean  
Dim value As System.Boolean  
  
value = instance.IsHidden(ConsiderSuppressed)
```

C#

```
System.bool IsHidden(  
    System.bool ConsiderSuppressed  
)
```

C++/CLI

```
System.bool IsHidden(  
& System.bool ConsiderSuppressed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConsiderSuppressed
Controls whether suppressed components are hidden

Return Value

True or false (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::IsHidden](#).

Example

[Change Material Properties \(VBA\)](#)
[Get Transforms of Assembly Components \(VBA\)](#)
[Get Component State \(C#\)](#)
[Get Component State \(VB.NET\)](#)
[Get Component State \(VBA\)](#)

Remarks

The state of this component can vary based on the active configuration.

ConsiderSuppressed	Component	Component	IsHidden
True	Hidden	Unsuppressed	True
True	Hidden	Suppressed	True
True	Shown	Unsuppressed	False
True	Shown	Suppressed	True
False	Hidden	Unsuppressed	True
False	Hidden	Suppressed	True

False	Shown	Unsuppressed	False
False	Shown	Suppressed	False

NOTE: For [lightweight](#) components, IsHidden returns True if ConsiderSuppressed is True.

 See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::ISetVisibility Method](#)
[IComponent2::IsSuppressed Method](#)
[IComponent2::SetSuppression2 Method](#)
[IComponent2::SetVisibility Method](#)
[IComponent2::Visible Property](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IsLoaded Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsLoaded Method (IComponent2)

Gets whether a component is loaded.

 .NET Syntax

Visual Basic (Declaration)

Function IsLoaded() As System.Boolean

Visual Basic (Usage)

```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.IsLoaded()
```

C#

System.bool IsLoaded()

C++/CLI

System.bool IsLoaded();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the component is loaded, false if not

 Visual Basic for Applications (VBA) Syntax

See [Component2::IsLoaded](#).

 Example

[Get Whether Components Are Loaded \(C#\)](#)
[Get Whether Components Are Loaded \(VB.NET\)](#)
[Get Whether Components Are Loaded \(VBA\)](#)

 See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2009 SP3, Revision Number 17.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsMirrored Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsMirrored Method (IComponent2)

Gets whether this component is mirrored.

.NET Syntax

Visual Basic (Declaration)

```
Function IsMirrored() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.IsMirrored()
```

C#

```
System.bool IsMirrored()
```

C++/CLI

```
System.bool IsMirrored();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the component is mirrored, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsMirrored](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IPartDoc::IsMirrored Method](#)
[IAssemblyDoc::MirrorComponents3 Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsNameOverridden Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsNameOverridden Method (IComponent2)

Gets whether this component's name is overridden.

.NET Syntax

Visual Basic (Declaration)

```
Function IsNameOverridden() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Boolean

value = instance.IsNameOverridden()
```

C#

```
System.bool IsNameOverridden()
```

C++/CLI

```
System.bool IsNameOverridden();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if component name is overridden, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsNameOverridden](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2022 SP02, Revision Number 30.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsPatternInstance Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsPatternInstance Method (IComponent2)

Gets whether the component is a member of a pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function IsPatternInstance() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Boolean

value = instance.IsPatternInstance()
```

C#

```
System.bool IsPatternInstance()
```

C++/CLI

```
System.bool IsPatternInstance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the component is a member of pattern instance, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsPatternInstance](#).

Remarks

True if the component is a member of pattern instance, false if not

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsRoot Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsRoot Method (IComponent2)

Gets whether this component is the root component.

.NET Syntax

Visual Basic (Declaration)

```
Function IsRoot() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Boolean

value = instance.IsRoot()
```

C#

```
System.bool IsRoot()
```

C++/CLI

```
System.bool IsRoot();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component is the root component, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsRoot](#).

Example

[Open Assembly in Large Design Review Mode \(VBA\)](#)
[Open Assembly in Large Design Review Mode \(VB.NET\)](#)
[Open Assembly in Large Design Review Mode \(C#\)](#)

Remarks

The root component is located at the top of the FeatureManager design tree in an assembly.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetModelDoc2 Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsSmartComponent Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsSmartComponent Method (IComponent2)

Gets whether this component is a Smart Component.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSmartComponent() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean
```

```
value = instance.IsSmartComponent()
```

C#

```
System.bool IsSmartComponent()
```

C++/CLI

```
System.bool IsSmartComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if a Smart Component, false if not

Example

[Activate Smart Features in an Assembly \(C#\)](#)
[Activate Smart Features in an Assembly \(VB.NET\)](#)
[Activate Smart Features in an Assembly \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetSmartComponentData Method](#)
[IComponent2::SetSmartComponentData Method](#)
[ISmartComponentFeatureData Interface](#)
[IAssemblyDoc::CreateSmartComponent Method](#)
[IAssemblyDoc::AddSmartComponent Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsSpeedPak Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsSpeedPak Property (IComponent2)

Gets whether the active configuration for this component is SpeedPak.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IsSpeedPak As System.Boolean

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean
```

```
value = instance.IsSpeedPak
```

C#

```
System.bool IsSpeedPak {get;}
```

C++/CLI

```
property System.bool IsSpeedPak {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the active configuration for this component is SpeedPak, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsSpeedPak](#).

Example

[Create SpeedPak for Subassemblies \(C#\)](#)
[Create SpeedPak for Subassemblies \(VB.NET\)](#)
[Create SpeedPak for Subassemblies \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CreateSpeedPak Method \(\)](#)
[IAssemblyDoc::SetSpeedPakConfigurations Method \(\)](#)
[IAssemblyDoc::SetSpeedPakToParent Method \(\)](#)
[IAssemblyDoc::UpdateSpeedPak Method \(\)](#)
[IAssemblyDoc::UseSpeedPak Method \(\)](#)
[IConfiguration::IsSpeedPak Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

[SOLIDWORKS API Help](#)

[Send comments on this topic.](#)

IsSuppressed Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsSuppressed Method (IComponent2)

Gets whether this component is suppressed.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSuppressed() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean
```

```
value = instance.IsSuppressed()
```

C#

```
System.bool IsSuppressed()
```

C++/CLI

```
System.bool IsSuppressed();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component is suppressed, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsSuppressed](#).

Example

[Change Material Properties \(VBA\)](#)
[Get and Set Component's Suppression State \(VBA\)](#)
[Get Transforms of Assembly Components \(VBA\)](#)
[Make Assembly Components Lightweight \(VBA\)](#)

Remarks

The suppression state of a component can vary based on the active configuration. You might want to use [IComponent2::GetSuppression](#), which returns the specific suppression state of the component.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::LightweightAllResolved Method](#)
[IAssemblyDoc::SetComponentSuppression Method](#)
[IComponent2::GetSuppression Method](#)
[IComponent2::SetSuppression2 Method](#)
[IFeature::IsSuppressed2 Method](#)
[IFeature::SetSuppression2 Method](#)
[IFeature::ISetSuppression2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsVirtual Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IsVirtual Property (IComponent2)

Gets whether this component is a virtual component.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Property IsVirtual As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
instance.IsVirtual = value  
  
value = instance.IsVirtual
```

C#

```
System.bool IsVirtual {get; set;}
```

C++/CLI

```
property System.bool IsVirtual {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the component is a virtual component, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::IsVirtual](#).

Example

[Insert New Virtual Component \(VB.NET\)](#)
[Insert New Virtual Component \(C#\)](#)
[Insert New Virtual Component \(VBA\)](#)

Remarks

When you create components in the context of an assembly, the software can save them inside the assembly file, and you can immediately begin modeling. Later, you can save the components to external files or delete them. Also, no **In-Place** mate is created, so you can position and constrain the component however you want.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method](#)
[IAssemblyDoc::InsertNewAssembly Method](#)
[IComponent2::SaveVirtualComponent Method](#)
[IComponent2::MakeVirtual Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ListExternalFileReferences Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ListExternalFileReferences Method (IComponent2)

Obsolete. Superseded by [IComponent2::ListExternalFileReferences2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub ListExternalFileReferences( _  
    ByRef ModelPathName As System.Object, _  
    ByRef ComponentPathName As System.Object, _  
    ByRef Feature As System.Object, _  
    ByRef DataType As System.Object, _  
    ByRef Status As System.Object, _  
    ByRef RefEntity As System.Object, _  
    ByRef FeatCom As System.Object _  
)
```

Visual Basic (Usage)

```
Dim instance As IComponent2  
Dim ModelPathName As System.Object  
Dim ComponentPathName As System.Object
```



```
Dim Feature As System.Object
Dim DataType As System.Object
Dim Status As System.Object
Dim RefEntity As System.Object
Dim FeatCom As System.Object
```

```
instance.ListExternalFileReferences(ModelPathName, ComponentPathName, Feature, DataType, Status,
RefEntity, FeatCom)
```

C#

```
void ListExternalFileReferences(
    out System.object ModelPathName,
    out System.object ComponentPathName,
    out System.object Feature,
    out System.object DataType,
    out System.object Status,
    out System.object RefEntity,
    out System.object FeatCom
)
```

C++/CLI

```
void ListExternalFileReferences(
    & [Out] System.Object^ ModelPathName,
    & [Out] System.Object^ ComponentPathName,
    & [Out] System.Object^ Feature,
    & [Out] System.Object^ DataType,
    & [Out] System.Object^ Status,
    & [Out] System.Object^ RefEntity,
    & [Out] System.Object^ FeatCom
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModelPathName

ComponentPathName

Feature

DataType

Status

RefEntity

FeatCom

Visual Basic for Applications (VBA) Syntax

See [Component2::ListExternalFileReferences](#).

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ListExternalFileReferences2 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ListExternalFileReferences2 Method (IComponent2)

Gets the names and statuses of the external file references on the component.

.NET Syntax

Visual Basic (Declaration)

```
Sub ListExternalFileReferences2( _
    ByRef ModelPathName As System.Object, _
```

```

ByRef ComponentPathName As System.Object, _
ByRef Feature As System.Object, _
ByRef DataType As System.Object, _
ByRef Status As System.Object, _
ByRef RefEntity As System.Object, _
ByRef FeatCom As System.Object, _
ByRef ConfigOption As System.Integer, _
ByRef ConfigName As System.String _
)

```

Visual Basic (Usage)



```

Dim instance As IComponent2
Dim ModelPathName As System.Object
Dim ComponentPathName As System.Object
Dim Feature As System.Object
Dim DataType As System.Object
Dim Status As System.Object
Dim RefEntity As System.Object
Dim FeatCom As System.Object
Dim ConfigOption As System.Integer
Dim ConfigName As System.String

```

```

instance.ListExternalFileReferences2(ModelPathName, ComponentPathName, Feature, DataType, Status,
RefEntity, FeatCom, ConfigOption, ConfigName)

```

C#

```

void ListExternalFileReferences2(
    out System.object ModelPathName,
    out System.object ComponentPathName,
    out System.object Feature,
    out System.object DataType,
    out System.object Status,
    out System.object RefEntity,
    out System.object FeatCom,
    out System.int ConfigOption,
    out System.string ConfigName
)

```

C++/CLI

```

void ListExternalFileReferences2(
& [Out] System.Object^ ModelPathName,
& [Out] System.Object^ ComponentPathName,
& [Out] System.Object^ Feature,
& [Out] System.Object^ DataType,
& [Out] System.Object^ Status,
& [Out] System.Object^ RefEntity,
& [Out] System.Object^ FeatCom,
& [Out] System.int ConfigOption,
& [Out] System.String^ ConfigName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModelPathName

Array of path names of documents

ComponentPathName

Array of path names of referenced components

Feature

Array of in-context items (sketches, features, and so on)

DataType

Array of data used to create the items (converted edge or face, converted or offset sketch entity, body, and so on)

Status

Array of the statuses of the external references as defined in [swExternalReferenceStatus_e](#)

RefEntity

Array of actual items being used and the names of the documents that contain the items

FeatCom

Array of the names of the components in which the affected features exist; this information is only displayed when one or more RefEntity is in a different component in an assembly and does not apply to derived parts

ConfigOption

Configuration options as defined by [swExternalFileReferencesConfig_e](#)

ConfigName

Name of the configuration when configOption is swExternalFileReferencesNamedConfig

Visual Basic for Applications (VBA) Syntax

See [Component2::ListExternalFileReferences2](#).

Example

[Get External References \(C#\)](#)
[Get External References \(VB.NET\)](#)
[Get External References \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IListExternalFileReferences2 Method](#)
[IComponent2::ListExternalFileReferencesCount Method](#)
[IComponent2::UpdateExternalFileReferences Method](#)
[IFeature::IListExternalFileReferences2 Method](#)
[IFeature::ListExternalFileReferences2 Method](#)
[IFeature::ListExternalFileReferencesCount Method](#)
[IFeature::UpdateExternalFileReferences Method](#)
[IModelDocExtension::IListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ListExternalFileReferencesCount Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ListExternalFileReferencesCount Method (IComponent2)

Gets the number of external references on the component.

.NET Syntax

Visual Basic (Declaration)

```
Function ListExternalFileReferencesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer  
  
value = instance.ListExternalFileReferencesCount()
```

C#

```
System.int ListExternalFileReferencesCount()
```

C++/CLI

```
System.int ListExternalFileReferencesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of external references

Visual Basic for Applications (VBA) Syntax

See [Component2::ListExternalFileReferencesCount](#).

Example

[Get External References \(VBA\)](#)
[Get External References \(VB.NET\)](#)
[Get External References \(C#\)](#)

Remarks

Call this method before calling [IComponent2::IListExternalFileReferences2](#) to get the number of external references of the component.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IListExternalFileReferences2 Method](#)
[IFeature::IListExternalFileReferences2 Method](#)
[IFeature::IListExternalFileReferences2 Method](#)
[UpdateExternalFileReferences Method](#)
[IComponent2::UpdateExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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MakeVirtual Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : MakeVirtual Method (IComponent2)

Obsolete. by Superseded by [IComponent2::MakeVirtual2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function MakeVirtual() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.MakeVirtual()
```

C#

```
System.bool MakeVirtual()
```

C++/CLI

```
System.bool MakeVirtual();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component is saved in an assembly, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::MakeVirtual](#).

Remarks

This method breaks the link to the external component file. Existing references are ignored, and the component is renamed.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

MakeVirtual2 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : MakeVirtual2 Method (IComponent2)

Makes this component and optionally its child components virtual by saving them in the current assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeVirtual2( _  
    ByVal AlsoChildVirtual As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AlsoChildVirtual As System.Boolean  
Dim value As System.Boolean  
  
value = instance.MakeVirtual2(AlsoChildVirtual)
```

C#

```
System.bool MakeVirtual2(  
    System.bool AlsoChildVirtual  
)
```

C++/CLI

```
System.bool MakeVirtual2(  
    & System.bool AlsoChildVirtual  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AlsoChildVirtual

True to make child components of this component virtual, false to keep the child components linked to external files

Return Value

True if this component and optionally its child components are saved in an assembly, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::MakeVirtual2](#).

Example

[Add Component and Mate \(C#\)](#)
[Add Component and Mate \(VB.NET\)](#)
[Add Component and Mate \(VBA\)](#)

Remarks

This method breaks the link to the external component file and optionally the links to any of its child component files. Existing references are ignored, and the component and optionally any of its child components are renamed.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IsVirtual Property \(\)](#)
[IComponent2::SaveVirtualComponent Method \(\)](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method \(\)](#)
[IAssemblyDoc::InsertNewVirtualPart Method \(\)](#)
[IModelDocExtension::IsVirtualComponent3 Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MaterialPropertyValues Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : MaterialPropertyValues Property (IComponent2)

Gets or sets the material properties for the selected component in the active configuration.

.NET Syntax

Visual Basic (Declaration)

Property MaterialPropertyValues As System.Object

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Object
```

```
instance.MaterialPropertyValues = value
```

```
value = instance.MaterialPropertyValues
```

C#

```
System.object MaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.Object^ MaterialPropertyValues {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles that describes the material values on this component (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::MaterialPropertyValues](#).

Example

[Change Material Properties \(VBA\)](#)
[Get Material Properties \(VBA\)](#)

Remarks

The material values include the component color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission. Valid values are from 0 to 1 for all values.

The format of the parameters or return values is an array of doubles as follows:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

This method returns a NULL VARIANT if the user has not explicitly modified this component from the material property values of the underlying part document. For example, the user can right-click a component in the FeatureManager design tree and select **Component Properties** to change the color. If the user has not done this, then this method returns NULL color information.

The default component color can be obtained from the component's [IModelDoc2](#) object ([IComponent2::GetModelIDoc](#)) using [IModelDoc2::MaterialPropertyValues](#).

You might also want to check for specific face colors using [IFace2::MaterialPropertyValues](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IMaterialPropertyValues Property](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::IMaterialPropertyValues2 Method](#)
[IComponent2::IRemoveMaterialProperty2 Method](#)
[IComponent2::RemoveMaterialProperty2 Method](#)
[IComponent2::ISetMaterialPropertyValues2 Method](#)
[IComponent2::SetMaterialPropertyValues2 Method](#)
[IComponent2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Name Property (IComponent2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IComponent2 Interface](#) : Name Property (IComponent2)

Obsolete. Superseded by [IComponent2::Name2](#).

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Name As System.String

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.Name
```

C#

```
System.string Name {get;}
```

C++/CLI

```
property System.String^ Name {
    System.String^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Component2::Name](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Name2 Property (IComponent2)

[See Also](#) [Example](#)



Gets or sets the name of the selected component.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property Name2 As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.String
```

```
instance.Name2 = value
```

```
value = instance.Name2
```

C#

```
System.string Name2 {get; set;}
```

C++/CLI

```
property System.String^ Name2 {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of this component

Visual Basic for Applications (VBA) Syntax

See [Component2::Name2](#).

Example

[Add Component and Mate \(C++\)](#)
[Change Material Properties \(VBA\)](#)
[Check Interference Using AssemblyDoc::ToolsCheckInterference2 \(VBA\)](#)
[Get Component Names and Types for Inplace Mate \(VBA\)](#)
[Get Component via Display Dimension \(VBA\)](#)
[Traverse Assembly at Component Level \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VB.NET\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(C#\)](#)
[Change Name of Component \(C#\)](#)
[Change Name of Component \(VB.NET\)](#)
[Change Name of Component \(VBA\)](#)

Remarks

This property returns a name that includes an instance number. For example:

Part1-1

indicates that this is the first instance of the Part1 Component2. If you are examining a component that is within a subassembly, then this property returns a name that includes the full hierarchical path of component names. For example:

subAssem1-2/Part1-1

indicates that this component (Part1) is the first instance within the subAssem1 Component2. It also shows that the second instance of subAssem1 is referenced.

If you are setting the name of a component:

- Before executing a name change, this property checks the swExtRefUpdateCompNames setting. If swExtRefUpdateCompNames is true, then the name change fails; if swExtRefUpdateCompNames is false, then the name change succeeds. Use [ISldWorks::GetUserPreferenceToggle](#) to change the swExtRefUpdateCompNames setting. Also, remember that some special characters are reserved by SOLIDWORKS, so be sure to use valid characters in the new name.
- The component must be selected.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetSelectByIdString Method](#)
[IAssemblyDoc::GetComponentByName Method](#)
[IAssemblyDoc::GetUnloadedComponentNames Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PresentationTransform Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : PresentationTransform Property (IComponent2)

Gets or sets the component transform.

.NET Syntax

Visual Basic (Declaration)

Property PresentationTransform As MathTransform

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As MathTransform  
  
instance.PresentationTransform = value  
value = instance.PresentationTransform
```

C#

```
MathTransform PresentationTransform {get; set;}
```

C++/CLI

```
property MathTransform^ PresentationTransform {  
    MathTransform^ get();  
    void set ( & MathTransform^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Component transform](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::PresentationTransform](#).

Example

[Use Presentation Transforms to Move Component \(VBA\)](#)

Remarks

This property affects the graphical display; it does not affect the underlying model geometry.

This property:

- gets or sets the component transform for graphical display.
- ignores all mate and in-context relationships. Only the display of the component on the screen is affected.
- does not apply the transform to component geometry.

• does not display any changes. To display changes, call [IModelDoc2::GraphicsRedraw2](#).
This property's functionality is for graphical purposes only and does not affect solid models. For example, if you want to animate the explode steps for an assembly, then maintaining assembly mate and in-context relationships is not needed or desirable.
You must first enable a presentation transform by setting [IAssemblyDoc::EnablePresentation](#) to true.
[IComponent2::RemovePresentationTransform](#) removes any transform applied by IComponent2::PresentationTransform. Set IAssemblyDoc::EnablePresentation to false after calling IComponent2::RemovePresentationTransform. After calling this method, the component is next drawn in a position consistent with its underlying geometry ([IComponent2::Transform2](#)).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetSpecificTransform Method \(\)](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReferencedConfiguration Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ReferencedConfiguration Property (IComponent2)

Gets or sets the active configuration used by this component.

.NET Syntax

Visual Basic (Declaration)

Property ReferencedConfiguration As System.String

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.String  
  
instance.ReferencedConfiguration = value  
value = instance.ReferencedConfiguration
```

C#

```
System.string ReferencedConfiguration {get; set;}
```

C++/CLI

```
property System.String^ ReferencedConfiguration {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of the configuration

Visual Basic for Applications (VBA) Syntax

See [Component2::ReferencedConfiguration](#).

Example

[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Get Transforms of Assembly Components \(VBA\)](#)
[Change Referenced Configuration \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VB.NET\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(C#\)](#)

Remarks

You can use this property to change the active configuration of:

- a pattern seed component.
- pattern instance components, but only when the pattern's ForceUseSeedConfiguration property is set to false.

When the configuration is changed, components might become suppressed or unsuppressed. This means that the array previously returned from [IComponent2::GetChildren](#) or [IComponent2::!GetChildren](#) becomes invalid. If an application is traversing an assembly and calls [IComponent2::ReferencedConfiguration](#), then it should stop the traversal and start again by getting the assembly's [IModelDoc2](#) and active configuration. The array previously returned from [IComponent2::GetChildren](#) or [IComponent2::!GetChildren](#) should be released before calling [IComponent2::ReferencedConfiguration](#). To hold onto the changed component, you can perform an extra [AddRef\(\)](#) on that component before freeing the array.

After changing the referenced configuration, you must call [IModelDoc2::EditRebuild3](#) to display the changes.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CompConfigProperties5 Method \(\)](#)
[IChainPatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[IDerivedPatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[ILocalCircularPatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[ILocalCurvePatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[ILocalLinearPatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[ILocalSketchPatternFeatureData::ForceUseSeedConfiguration Property \(\)](#)
[IMirrorComponentFeatureData::ForceUseSeedConfiguration Property \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReferencedDisplayState Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ReferencedDisplayState Property (IComponent2)

Obsolete. Superseded by [IComponent2::ReferencedDisplayState2](#).

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property ReferencedDisplayState As System.String

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.String
```

```
value = instance.ReferencedDisplayState
```

C#

```
System.string ReferencedDisplayState {get;}
```

C++/CLI

```
property System.String^ ReferencedDisplayState {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of display state

Visual Basic for Applications (VBA) Syntax

See [Component2::ReferencedDisplayState](#).

Example

[Get Referenced Display State \(VBA\)](#)
[Get Referenced Display State \(VB.NET\)](#)
[Get Referenced Display State \(C#\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IConfiguration::GetDisplayStates Method](#)
[IConfigurationManager::LinkDisplayStatesToConfigurations Property \(\)](#)

Availability

SOLIDWORKS 2011 SP02, Revision Number 19.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReferencedDisplayState2 Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ReferencedDisplayState2 Property (IComponent2)

Gets or sets the active display state of this component.

.NET Syntax

Visual Basic (Declaration)

Property ReferencedDisplayState2 As System.String

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.String
```

```
instance.ReferencedDisplayState2 = value
```

```
value = instance.ReferencedDisplayState2
```

C#

```
System.string ReferencedDisplayState2 {get; set;}
```

C++/CLI

```
property System.String^ ReferencedDisplayState2 {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of display state

Visual Basic for Applications (VBA) Syntax

See [Component2::ReferencedDisplayState2](#).

Example

[Set Active Display State of Referenced Component \(C#\)](#)
[Set Active Display State of Referenced Component \(VB.NET\)](#)
[Set Active Display State of Referenced Component \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IConfiguration::GetDisplayStates Method \(\)](#)
[IConfigurationManager::LinkDisplayStatesToConfigurations Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveMaterialProperty Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : RemoveMaterialProperty Method (IComponent2)

Obsolete. Superseded by [IComponent2::RemoveMaterialProperty2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveMaterialProperty() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean
```

```
value = instance.RemoveMaterialProperty()
```

C#

```
System.bool RemoveMaterialProperty()
```

C++/CLI

```
System.bool RemoveMaterialProperty();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Component2::RemoveMaterialProperty](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveMaterialProperty2 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : RemoveMaterialProperty2 Method (IComponent2)

Removes the appearance from the component.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveMaterialProperty2( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Boolean
```

```
value = instance.RemoveMaterialProperty2(Config_opt, Config_names)
```

C#

```
System.bool RemoveMaterialProperty2(  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.bool RemoveMaterialProperty2(  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration options as defined by [swInConfigurationOpts_e](#)

Config_names

Array of configuration names for this component

Return Value

True if the component's appearance is removed, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::RemoveMaterialProperty2](#).

Example

[Remove Material Properties from Assembly Component \(VBA\)](#)

Remarks

This method is intended to be used on a component whose appearance has changed; for example, color.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::IRemoveMaterialProperty2 Method](#)

[IComponent2::GetMaterialPropertyValues2 Method](#)

[IComponent2::IGetMaterialPropertyValues2 Method](#)

[IComponent2::ISetMaterialPropertyValues2 Method](#)

[IComponent2::SetMaterialPropertyValues2 Method](#)

[IComponent2::MaterialPropertyValues Property](#)

[IComponent2::IMaterialPropertyValues Property](#)

[IComponent2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemovePresentationTransform Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : RemovePresentationTransform Method (IComponent2)

Removes the presentation transform from this component.

.NET Syntax

Visual Basic (Declaration)

```
Sub RemovePresentationTransform()
```

```
Dim instance As IComponent2

instance.RemovePresentationTransform()
```

C#

```
void RemovePresentationTransform()
```

C++/CLI

```
void RemovePresentationTransform();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Component2::RemovePresentationTransform](#).

Example

Use Presentation Transforms to Move Component ([VBA](#)).

Remarks

Set [IAssemblyDoc::EnablePresentation](#) to false after a user is finished with presentation transforms and has called [IComponent2::RemovePresentationTransform](#), which removes any transform applied by [IComponent::PresentationTransform](#).

After calling [IComponent2::RemovePresentationTransform](#), the component is next drawn in a position consistent with its underlying geometry ([IComponent2::Transform2](#)).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::EnablePresentation Property](#)
[IComponent2::GetTotalTransform Method](#)
[IComponent2::PresentationTransform Property](#)
[IComponent2::Transform2 Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

RemoveTexture Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : RemoveTexture Method (IComponent2)

Removes the texture from this component in the specified configuration.

I.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTexture( _
    ByVal Config\_name As System.String _
) As System.Boolean
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IComponent2
Dim Config_name As System.String
Dim value As System.Boolean

value = instance.RemoveTexture(Config_name)
```

C#

```
System.bool RemoveTexture(  
    System.string Config\_name  
)
```

C++/CLI

```
System.bool RemoveTexture(  
& System.String^ Config\_name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_name
Name of the configuration

Return Value

True if texture is removed, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::RemoveTexture](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTexture Method](#)
[IComponent2::SetTexture Method](#)
[IComponent2::RemoveTextureByDisplayState Method](#)
[IComponent2::SetTextureByDisplayState Method](#)
[IModelDocExtension::CreateTexture Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveTextureByDisplayState Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : RemoveTextureByDisplayState Method (IComponent2)

Removes the texture applied to this component in the specified display state.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTextureByDisplayState(  
    ByVal DisplayStateName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim DisplayStateName As System.String  
Dim value As System.Boolean  
  
value = instance.RemoveTextureByDisplayState(DisplayStateName)
```

C#

```
System.bool RemoveTextureByDisplayState(  
    System.string DisplayStateName  
)
```

C++/CLI

```
System.bool RemoveTextureByDisplayState(  
& System.String^ DisplayStateName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayStateName
Display state name

Return Value

True if the texture is removed from the component, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::RemoveTextureByDisplayState](#).

Example

[Apply and Remove Texture To and From Component By Display State \(C#\)](#)
[Apply and Remove Texture To and From Component By Display State \(VB.NET\)](#)
[Apply and Remove Texture To and From Component By Display State \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTexture Method](#)
[IComponent2::RemoveTexture Method](#)
[IComponent2::SetTexture Method](#)
[IComponent2::SetTextureByDisplayState Method](#)
[IModelDocExtension::CreateTexture Method](#)

Availability

SOLIDWORKS 2010 SP1, Revision Number 18.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReplaceReference Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ReplaceReference Method (IComponent2)

Obsolete. Superseded by [AssemblyDoc::ReplaceComponents](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ReplaceReference( _  
    ByVal FileName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim FileName As System.String  
Dim value As System.Integer  
  
value = instance.ReplaceReference(FileName)
```

C#

```
System.int ReplaceReference(  
    System.string FileName  
)
```

C++/CLI

```
System.int ReplaceReference(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Visual Basic for Applications (VBA) Syntax

See [Component2::ReplaceReference](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ResetPropertyExtension Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ResetPropertyExtension Method (IComponent2)

Clears all of the values stored in the property extension.

.NET Syntax

Visual Basic (Declaration)

```
Sub ResetPropertyExtension()
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
instance.ResetPropertyExtension()
```

C#

```
void ResetPropertyExtension()
```

C++/CLI

```
void ResetPropertyExtension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Component2::ResetPropertyExtension](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::AddPropertyExtension Method](#)
[IComponent2::GetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SaveVirtualComponent Method (IComponent2)

[See Also](#) [Example](#)



Saves a virtual component to an external file.

■ .NET Syntax

Visual Basic (Declaration)

```
Function SaveVirtualComponent( _  
    ByVal CompPathName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim CompPathName As System.String  
Dim value As System.Boolean  
  
value = instance.SaveVirtualComponent(CompPathName)
```

C#

```
System.bool SaveVirtualComponent(  
    System.string CompPathName  
)
```

C++/CLI

```
System.bool SaveVirtualComponent(  
& System.String^ CompPathName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompPathName
String containing full pathname of file to save component to

Return Value

True if successful, false if not

■ Visual Basic for Applications (VBA) Syntax

See [Component2::SaveVirtualComponent](#).

■ Example

[Insert New Virtual Component \(C#\)](#)
[Insert New Virtual Component \(VB.NET\)](#)
[Insert New Virtual Component \(VBA\)](#)

■ See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IsVirtual Property](#)
[IModelDocExtension::IsVirtualComponent3 Method](#)
[IAssemblyDoc::InsertNewVirtualPart Method](#)
[IAssemblyDoc::InsertNewAssembly Method](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method](#)
[IComponent2::MakeVirtual Method](#)

■ Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0



Obsolete. Superseded by [IComponent2::Select3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select( _  
    ByVal AppendFlag As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AppendFlag As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.Select(AppendFlag)
```

C#

```
System.bool Select(  
    System.bool AppendFlag  
)
```

C++/CLI

```
System.bool Select(  
& System.bool AppendFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppendFlag

Visual Basic for Applications (VBA) Syntax

See [Component2::Select](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select2 Method (IComponent2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IComponent2 Interface](#) : Select2 Method (IComponent2)

Obsolete. Superseded by [IComponent2::Select3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select2( _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Append As System.Boolean  
Dim Mark As System.Integer
```

```
Dim value As System.Boolean

value = instance.Select2(Append, Mark)
```

C#

```
System.bool Select2(
    System.bool Append,
    System.int Mark
)
```

C++/CLI

```
System.bool Select2(
    & System.bool Append,
    & System.int Mark
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

Mark

Visual Basic for Applications (VBA) Syntax

See [Component2::Select2](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select3 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : Select3 Method (IComponent2)

Obsolete. Superseded by [IComponent2::Select4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select3( _
    ByVal Append As System.Boolean, _
    ByVal Data As SelectData _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim Append As System.Boolean
Dim Data As SelectData
Dim value As System.Boolean

value = instance.Select3(Append, Data)
```

C#

```
System.bool Select3(
    System.bool Append,
    SelectData Data
)
```

C++/CLI

```
System.bool Select3(
& System.bool Append,
& SelectData^ Data
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

True appends the selection to the selection list, false replaces the selection list

Data

Pointer to the [ISelectData](#) object

Return Value

True if the component is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::Select3](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select4 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : Select4 Method (IComponent2)

Selects this component.

.NET Syntax

Visual Basic (Declaration)

```
Function Select4( _
    ByVal Append As System.Boolean, _
    ByVal Data As SelectData, _
    ByVal ShowPopup As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim Append As System.Boolean
Dim Data As SelectData
Dim ShowPopup As System.Boolean
Dim value As System.Boolean
```

```
value = instance.Select4(Append, Data, ShowPopup)
```

C#

```
System.bool Select4(
    System.bool Append,
    SelectData Data,
    System.bool ShowPopup
)
```

C++/CLI

```
System.bool Select4(
& System.bool Append,
```

```
& SelectData^ Data,  
& System.bool ShowPopup  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

True appends the selection to the selection list, false replaces the selection list

Data

Pointer to the [ISelectData](#) object

ShowPopup

True to show shortcut menu, false to not

Return Value

True if the component is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::Select4](#).

Example

[Fire Notifications When Renaming Components \(C#\)](#)

[Fire Notifications When Renaming Components \(VB.NET\)](#)

[Fire Notifications When Renaming Components \(VBA\)](#)

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[ISelectionMgr::GetSelectedObjectsComponent3 Method](#)

[IModelDocExtension::SelectAll Method](#)

[IComponent2::DeSelect Method \(\)](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17

SOLIDWORKS API Help

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SelectByMark Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SelectByMark Method (IComponent2)

Obsolete. Superseded by [IComponent2::Select3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SelectByMark( _  
    ByVal AppendFlag As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AppendFlag As System.Boolean  
Dim Mark As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SelectByMark(AppendFlag, Mark)
```

C#

```
System.bool SelectByMark(  
    System.bool AppendFlag,  
    System.int Mark  
)
```

C++/CLI

```
System.bool SelectByMark(  
& System.bool AppendFlag,  
& System.int Mark  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppendFlag

Mark

Visual Basic for Applications (VBA) Syntax

See [Component2::SelectByMark](#).

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetCosmosWorksMaterial Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SetCosmosWorksMaterial Method (IComponent2)

Assigns the material to use during analysis to this component.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCosmosWorksMaterial( _  
    ByVal ConfigName As System.String, _  
    ByVal Type As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ConfigName As System.String  
Dim Type As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SetCosmosWorksMaterial(ConfigName, Type)
```

C#

```
System.bool SetCosmosWorksMaterial(  
    System.string ConfigName,  
    System.int Type  
)
```

C++/CLI

```
System.bool SetCosmosWorksMaterial(  
& System.String^ ConfigName,  
& System.int Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName

Name of the configuration

Type

Type of material to assign as defined by [swCosmosWorksMat](#)

Return Value

True if the material is assigned, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::SetCosmosWorksMaterial](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IPartDoc::SetCosmosWorksMaterial Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetExcludeFromBOM2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SetExcludeFromBOM2 Method (IComponent2)

Sets whether to exclude this component from the bills of materials (BOMs) in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function SetExcludeFromBOM2( _  
    ByVal Exclude As System.Boolean, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Exclude As System.Boolean  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Integer  
  
value = instance.SetExcludeFromBOM2(Exclude, Config_opt, Config_names)
```

C#

```
System.int SetExcludeFromBOM2(  
    System.bool Exclude,  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.int SetExcludeFromBOM2(  
& System.bool Exclude,  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Exclude

True to exclude it, false to not

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of the names of configurations for which to set Exclude; null or Nothing if Config_opt is set to swInConfigurationOpts.swAllConfiguration or swInConfigurationOpts.swThisConfiguration

Return Value

Return code as defined in [swExcludeFromBOMError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::SetExcludeFromBOM2](#).

Remarks

This method is valid only for [table-based bills of materials](#); it does not work for Microsoft Excel-based bills of materials.

If you set Exclude to true, the name of the component changes in the FeatureManager design tree of the specified configuration; **(Excluded from BOM)** is appended. To update the FeatureManager design tree, call [IFeatureManager::UpdateFeatureTree](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CompConfigProperties6 Method \(\)](#)
[IComponent2::GetExcludeFromBOM2 Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetMaterialIdName Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SetMaterialIdName Method (IComponent2)

Sets the material name for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function SetMaterialIdName( _  
    ByVal Name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Name As System.String  
Dim value As System.Boolean  
  
value = instance.SetMaterialIdName(Name)
```

C#

```
System.bool SetMaterialIdName(  
    System.string Name  
)
```

C++/CLI

```
System.bool SetMaterialIdName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Material name

Return Value

True if the material name was set, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::SetMaterialIdName](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetMaterialIdName Method](#)
[IComponent2::SetMaterialUserName Method](#)
[IComponent2::GetMaterialUserName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetMaterialPropertyValues2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SetMaterialPropertyValues2 Method (IComponent2)

Sets the material properties for this component.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetMaterialPropertyValues2( _  
    ByVal Material\_values As System.Object, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Material_values As System.Object  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
  
instance.SetMaterialPropertyValues2(Material_values, Config_opt, Config_names)
```

C#

```
void SetMaterialPropertyValues2(  
    System.object Material\_values,  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
void SetMaterialPropertyValues2(  
& System.Object^ Material\_values,  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Material_values

Array of material values for this component (see **Remarks**)

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names for this component

Visual Basic for Applications (VBA) Syntax

See [Component2::SetMaterialPropertyValues2](#).

Remarks

The material property values include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

The format of Material_values is an array of doubles:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after calling this method.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::SetMaterialPropertyValues2 Method](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::RemoveMaterialProperty2 Method](#)
[IComponent2::RemoveMaterialProperty2 Method](#)
[IComponent2::MaterialPropertyValues Property](#)
[IComponent2::MaterialPropertyValues Property](#)
[IComponent2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetMaterialUserName Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : SetMaterialUserName Method (IComponent2)

Sets the material user name for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function SetMaterialUserName( _  
    ByVal Name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Name As System.String  
Dim value As System.Boolean  
  
value = instance.SetMaterialUserName(Name)
```

C#

```
System.bool SetMaterialUserName(  
    System.string Name  
)
```

C++/CLI

```
System.bool SetMaterialUserName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Material user name property

Return Value

True if the material user name was set, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::SetMaterialUserName](#).

Remarks

This material name is visible to the user.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::GetMaterialUserName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0